

Elodie Barthout

As a millennial who grew up during the development and expansion of the internet as well as the arrival of smartphones, needless to say that I struggle to keep mine away. Always looking for it only to spend too much time dumbscrolling instead of enjoying boredom.

But, as someone who also feels overwhelmed when I receive too many notifications, I find myself putting my phone on the blissful “do not disturb” mode 90% of the time and even though I have a lot of hobbies outside of that evil machine (reading, sewing, playing video games...), I still spend quite some time on my phone on a daily basis.

So when I found that dataset on Kaggle, I thought it would be a perfect reminder of how bad this habit can be on our wellbeing, for me and perhaps you as well.

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Exploring the relationship between smartphone usage and quality of life

Dataset: Sleep, Screen Time and Stress Analysis available on Kaggle (CC0: Public Domain)
https://www.kaggle.com/datasets/jayjoshi37/sleep-screen-time-and-stress-analysis/data?select=sleep_mobile_stress_dataset_15000.csv

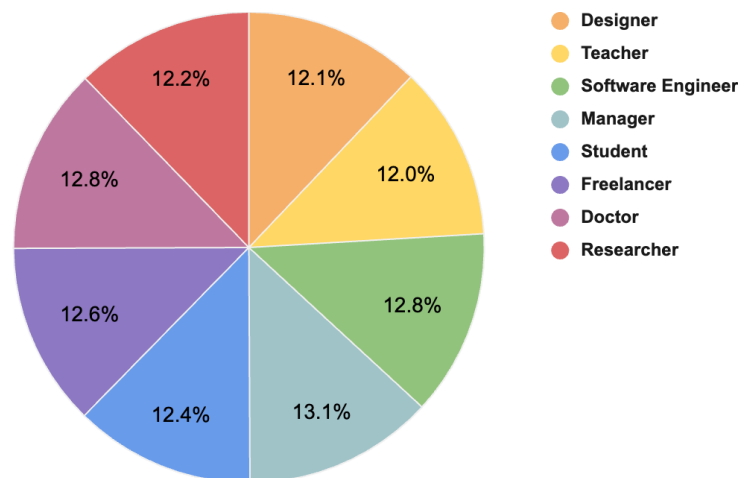
Last update in March 2026

I. How the time spent on mobile devices can affect our quality of life?

This dataset contains information about **15 000 people** from **18 to 59 years old** working in various fields: from students to teachers, designers, freelancers, doctors, managers, researchers & software engineers.

The goal of this analysis is to determine how our daily screen time can affect our quality of life by studying sleep patterns, stress level and mental fatigue scores.

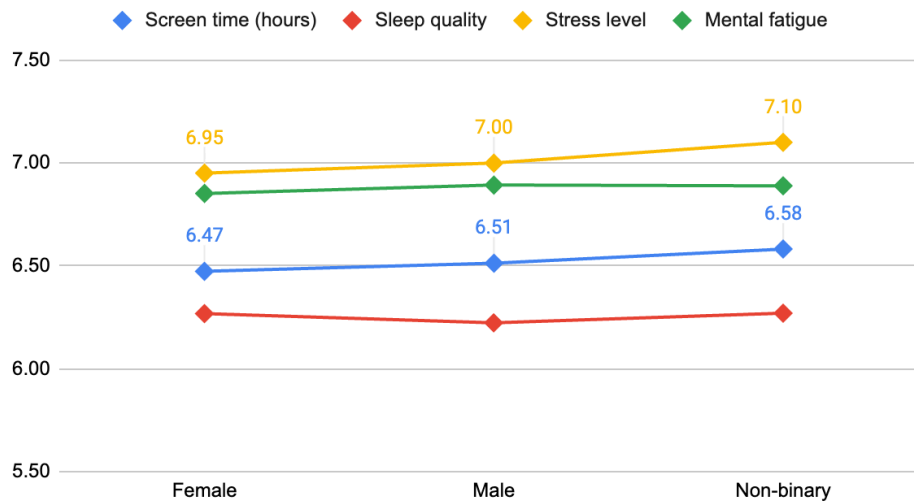
I started by checking if the data represented all genders and occupations equally and it was: 48% of females, 48% of males and 4% of non-binary people with all occupations represented in the same proportions:



I.1 Are there any differences between genders?

I started my exploration by analysing genders to see if we might behave differently on the matter of smartphone usage:

Impact of daily screen time on quality of life



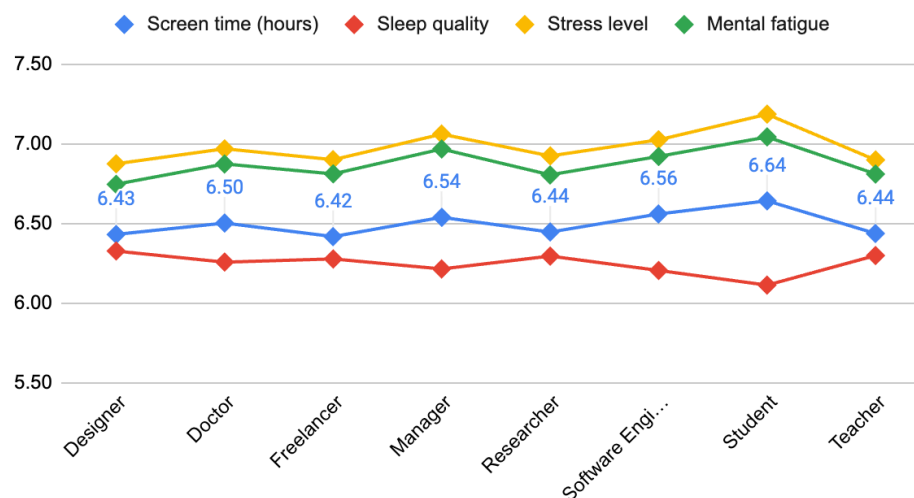
NB: sleep quality, mental fatigue and stress level being scored from 1 (low) to 10 (high)

There seems to be a correlation between the screen time and the stress level as the longer the screen time, the higher the stress level; but there are no differences concerning sleep quality or mental fatigue.

I.2 Are there any differences between professional occupations?

I then explored the professional occupations, assuming that maybe, some occupations would have a lower daily screen time than others (i.e Doctors):

Impact of daily screen time on quality of life

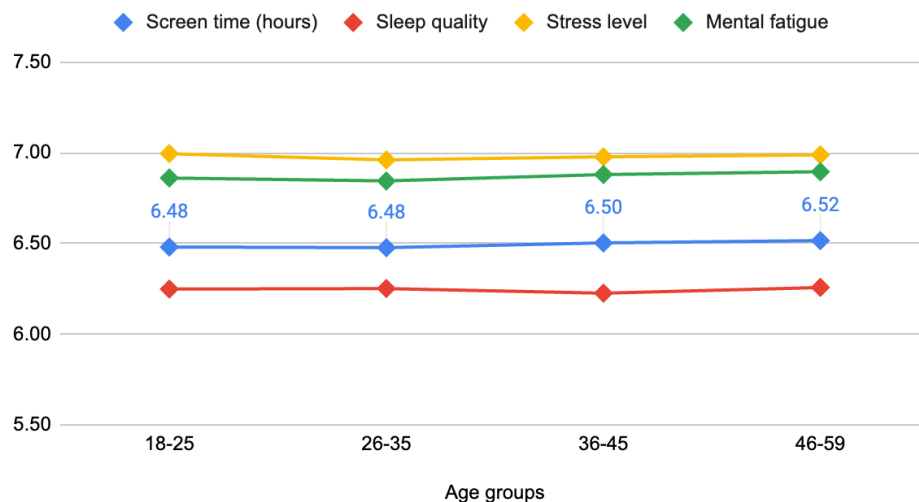


We can see that students have the longest screen time (and highest stress level) with 6,64h a day against 6,42h for the freelancers; but it is important to mention that the difference is only 13 min. There are also a lot of correlations here. First, the screen time seems to correlate with both stress level and mental fatigue, all evolving together. The screen time also seems to affect the quality of sleep, the latter being decreased when the former increased. Finally, since we can only determine correlations and not causations, stress level and mental fatigue are correlated together as well as sleep quality being correlated with both of these factors.

I.3 Are there any differences between age groups?

Next, I assumed that the daily screen time would vary quite a lot between age groups and with it, the quality of life perhaps:

Impact of daily screen time on quality of life



To my surprise, the values were pretty similar without any differences regarding any of the factors, suggesting that young people are not more glued to their phone as we might have thought at first.

I.4 If we combine all demographic information, are there any differences then?

Finally, I wondered what the data would show if I was combining all the information about gender, occupation & age group:

18-25 years old								
Female	Screen time		Sleep quality		Stress level		Mental fatigue	
	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest
	Student	Designer	Designer	Student	Student	Designer	Student	Designer
	$\Delta = 28$ min		$\Delta = 0.23$		$\Delta = 0.73$		$\Delta = 0.73$	
Male	Teacher	Designer	Designer	Teacher	Teacher	Designer	Teacher	Designer
	$\Delta = 37$ min		$\Delta = 0.25$		$\Delta = 0.56$		$\Delta = 0.49$	
Other	Student	Manager	Designer	Doctor	Student	Freelancer	Student	Freelancer
	$\Delta = 67$ min		$\Delta = 1.05$		$\Delta = 2.46$		$\Delta = 2.19$	
26-35 years old								
Female	Screen time		Sleep quality		Stress level		Mental fatigue	
	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest
	Software engineer	Freelancer	Freelancer	Manager	Manager	Freelancer	Manager	Designer
	$\Delta = 29$ min		$\Delta = 0.28$		$\Delta = 0.46$		$\Delta = 0.41$	
Male	Software engineer	Freelancer	Freelancer	Student	Software engineer	Freelancer	Software engineer	Freelancer
	$\Delta = 29$ min		$\Delta = 0.29$		$\Delta = 0.56$		$\Delta = 0.58$	
Non-binary	Student	Freelancer	Researcher	Student	Student	Researcher	Student	Freelancer
	$\Delta = 142$ min		$\Delta = 1.58$		$\Delta = 2.03$		$\Delta = 1.85$	
36-45 years old								
Female	Screen time		Sleep quality		Stress level		Mental fatigue	
	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest
	Designer	Teacher	Teacher	Manager	Doctor	Teacher	Doctor	Teacher
	$\Delta = 29$ min		$\Delta = 0.41$		$\Delta = 0.61$		$\Delta = 0.60$	
Male	Freelancer	Doctor	Doctor	Software engineer	Software engineer	Doctor	Manager	Doctor
	$\Delta = 41$ min		$\Delta = 0.29$		$\Delta = 0.55$		$\Delta = 0.54$	
Non-binary	Teacher	Designer	Designer	Doctor	Doctor	Designer	Freelancer	Designer
	$\Delta = 155$ min		$\Delta = 1.7$		$\Delta = 2.14$		$\Delta = 2.22$	
46-59 years old								
Female	Screen time		Sleep quality		Stress level		Mental fatigue	
	Highest	Lowest	Highest	Lowest	Highest	Lowest	Highest	Lowest
	Student	Teacher	Teacher	Student	Student	Teacher	Student	Teacher
	$\Delta = 18$ min		$\Delta = 0.28$		$\Delta = 0.39$		$\Delta = 0.34$	
Male	Manager	Designer	Designer	Student	Student	Designer	Student	Designer
	$\Delta = 31$ min		$\Delta = 0.50$		$\Delta = 0.64$		$\Delta = 0.58$	
Other	Researcher	Manager	Manager	Researcher	Designer	Teacher	Designer	Teacher
	$\Delta = 55$ min		$\Delta = 1.2$		$\Delta = 0.95$		$\Delta = 1.08$	

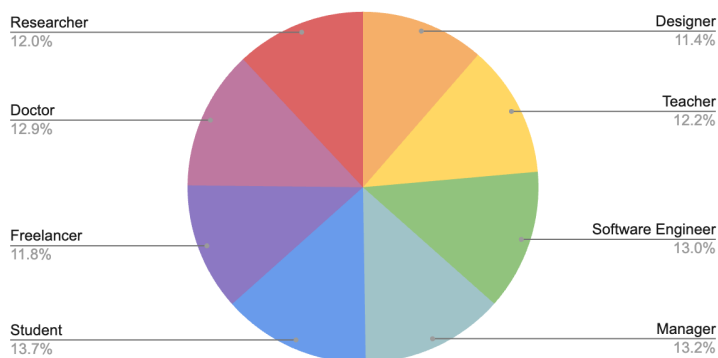
This table is a little bit crowded but basically, we can draw the same conclusions as the aforementioned graph about occupations. The longer the screen time, the higher the stress level and mental fatigue along with the lowest sleep quality; the reverse being true as well. Again, since we cannot determine causation, we can also mention other correlations such as stress level with mental fatigue, sleep quality along with both stress level and mental fatigue. All of this being true for all age groups and genders, all professional occupations being represented as well.

II. What about people with a stress level or mental fatigue score of 10?

Finally, I was curious about the scores ranging from 1 to 10 but I wondered if some people actually picked the extreme values and if there was anything to mention on that topic:

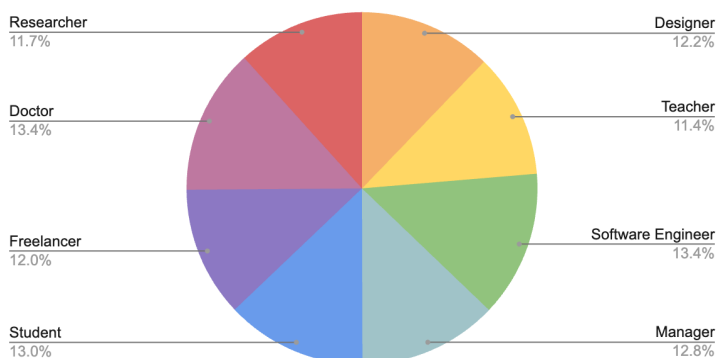
Percentage of occupations with a stress level score of 10

3875 people so 26% of the panel



Percentage of occupations with a mental fatigue score of 10

2484 people so 17% of the panel



A lot of people picked a score of 10 for both stress level and mental fatigue, 26% regarding stress level and 17% regarding mental fatigue. Interestingly, all professional occupations and genders were equally represented for each, suggesting that all professions come with their lot of stress, even the ones with more flexibility and/or freedom such as freelancers.

NB: this dataset also contained values regarding daily: physical activity, amount of notifications received, sleep duration, cups of caffeine; but I didn't see any trends so I didn't include them in this presentation.

III. Summary

This analysis aimed to determine if the daily screen time was affecting the quality of life. To answer that question, we had some demographic information along with data concerning sleep patterns, stress level and mental fatigue scores.

There were no differences concerning age groups. They were all sharing the same amount of daily screen time and the same lot of stress and mental fatigue. However, regarding genders, there was a correlation between screen time and stress level, both increasing together. Finally, by analysing occupations only as well as at the combination of all demographic factors; we found out that the screen time was negatively affecting the sleep quality, stress level and mental fatigue of the participants. Since the data collected does not allow us to determine causation effects, it also means that all factors are correlating together. This suggests that perhaps a poor quality of sleep is the leading cause of a higher stress level and mental fatigue, or that a higher stress level is leading to a poor sleep quality. This also suggests that a longer screen time might be the consequence of a high stress level or a high mental fatigue. We cannot know for sure, but it can only do us some good to reduce the time spent on our smartphones and connect with other human beings instead.

If you'd like to get to know the person behind this a little better

I am a Frenchie who moved to Sweden a few years ago to work as a Researcher. My first love being biology, understanding the subtleties of cancer to find ways to eliminate it is absolutely captivating. Obviously there's a big part of data analysis as a Researcher, & I got into computational work during my postdoctoral fellowship & found that so interesting and exciting that I wanted to dig more into it. That's how I ended up learning about data analytics in general & I really enjoyed it. I really like how my "natural talents" of overanalysing, overthinking and being detail-oriented, the same traits that are really helpful in research, are actually really helpful in data analytics too.

I spent the last few months getting trained on several tools, and here I am, writing those lines, showcasing one of my first data analysis work on spreadsheets, hoping you would consider hiring me (please do).

I hope this case study gave you some insights about my critical thinking strategy and just how my brain works in general. I hope you enjoyed it and are now thinking of interviewing me (I'd be thrilled). I know I can do a good job and of course there are obviously things I don't know yet, but I am eager to learn and have not a single doubt that I can, and will, learn anything there is to learn so please, take a chance on me (Abba fan, cliché I know).

Thank you so much for your time and see you hopefully very soon!

Elodie